

Take-Home Problem for Final Exam

March 22, 2025

This assignment is open book and open notes. The TAs may not provide any help. You will be graded subjectively on the strength, accuracy, and readability (including neatness) of your argument. Aim for a terse but complete exposition, and use the techniques and conventions as developed in the course. To discourage throwing everything at the wall and hoping for partial credit, I will deduct points for pointless diversions.

Warning: do not attempt to use internet solutions for this problem. It was chosen in part because several solutions to similar problems online have subtle mistakes that do not impact the final result, and which I will be alert for. ChatGPT's attempt at it was simply adorable. If you are caught cheating in this course, you will fail the entire course.

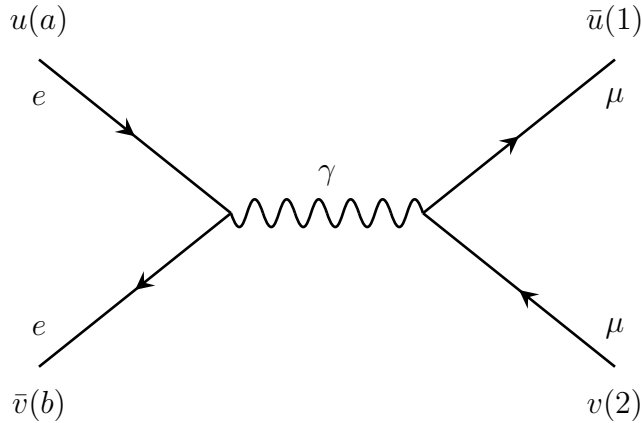


Figure 1: Leading order Feynman diagram for the process $e^- + e^+ \rightarrow \gamma \rightarrow \mu^- + \mu^+$. The particle plane-wave bi-spinor is u and the anti-particle plane-wave bi-spinor is v .

We will consider the process $e^- + e^+ \rightarrow \mu^- + \mu^+$ at leading order, as shown in Fig. 1. In Homework 7 you found the amplitude for this process to be:

$$\mathcal{M} = -\frac{g_e^2}{(p_a + p_b)^2} \bar{v}(b) \gamma^\mu u(a) \bar{u}(1) \gamma_\mu v(2)$$

where:

$$\gamma_\mu \equiv g_{\mu\nu} \gamma^\nu$$

In this problem, we will consider an experiment that is capable of supplying polarized beams, so that the incident electrons and positrons can be prepared with a known helicity. We will consider

the limit that $m_e, m_\mu \ll E$. In this limit the helicity eigenstates are identical to the chirality eigenstates, which for particles are:

$$u_R = N \begin{pmatrix} C \\ Se^{i\phi} \\ C \\ Se^{i\phi} \end{pmatrix}, \quad u_L = N \begin{pmatrix} -S \\ Ce^{i\phi} \\ S \\ -Ce^{i\phi} \end{pmatrix},$$

and for anti-particles are:

$$v_R = N \begin{pmatrix} S \\ -Ce^{i\phi} \\ -S \\ Ce^{i\phi} \end{pmatrix}, \quad v_L = N \begin{pmatrix} C \\ Se^{i\phi} \\ C \\ Se^{i\phi} \end{pmatrix},$$

with

$$C \equiv \cos\left(\frac{\theta}{2}\right), \quad S \equiv \sin\left(\frac{\theta}{2}\right)$$

and $N = \sqrt{E}$.

Part 1: For particle (a) take $\theta = 0$ and $\phi = 0$ and for anti-particle (b) take $\theta = \pi$ and $\phi = \pi$. Work out the chirality eigenstates for the particle (a) as:

$$u_R(a) = \sqrt{E} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_L(a) = \sqrt{E} \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix},$$

and for the anti-particle (b) as:

$$v_R(a) = \sqrt{E} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_L(a) = \sqrt{E} \begin{pmatrix} 0 \\ -1 \\ 0 \\ -1 \end{pmatrix},$$

Part 2: For particle (1) take $\phi = 0$ and work out the chirality eigenstates as:

$$u_R(1) = \sqrt{E} \begin{pmatrix} C \\ S \\ C \\ S \end{pmatrix}, \quad u_L(1) = \sqrt{E} \begin{pmatrix} -S \\ C \\ S \\ -C \end{pmatrix}$$

Part 3: For anti-particle (2) take $\phi = \pi$ and replace $\theta \rightarrow \pi - \theta$ to work out the chirality eigenstates:

$$v_R(2) = \sqrt{E} \begin{pmatrix} C \\ S \\ -C \\ -S \end{pmatrix}, \quad v_L(2) = \sqrt{E} \begin{pmatrix} S \\ -C \\ S \\ -C \end{pmatrix}$$

In the amplitude, we encounter vector currents of the form:

$$\begin{aligned} J^\mu &= \bar{v}_L(b) \gamma^\mu u_R(a) \\ P^\mu &= \bar{u}_R(1) \gamma^\mu v_L(2) \\ Q^\mu &= \bar{u}_L(1) \gamma^\mu v_R(2) \end{aligned}$$

Part 4: Calculate the currents J^μ , P^μ and Q^μ . Partial answer:

$$\begin{aligned} (J^0, J^1, J^2, J^3) &= 2E(0, -1, -i, 0) \\ (P^0, P^1, P^2, P^3) &= 2E(0, -\cos \theta, i, \sin \theta) \\ (Q^0, Q^1, Q^2, Q^3) &= ??? \end{aligned}$$

(To be clear: You must still provide the calculation even for the currents with an answer provided!)

Part 5: Consider the process with the specific chiralities $e^-(R) + e^+(L) \rightarrow \mu^-(R) + \mu^+(L)$ which has amplitude:

$$\mathcal{M}_{RL \rightarrow RL} = -\frac{g_e^2}{(p_a + p_b)^2} J^\mu P^\nu g_{\mu\nu}$$

Show that:

$$\mathcal{M}_{RL \rightarrow RL} = g_e^2(1 + \cos \theta)$$

Part 6: Consider the process with the specific chiralities $e^-(R) + e^+(L) \rightarrow \mu^-(L) + \mu^+(R)$ which has amplitude:

$$\mathcal{M}_{RL \rightarrow LR} = -\frac{g_e^2}{(p_a + p_b)^2} J^\mu Q^\nu g_{\mu\nu}$$

Calculate $\mathcal{M}_{RL \rightarrow LR}$ as in the previous part.

Part 7: Calculate the amplitude for the process $e^-(R) + e^+(L) \rightarrow \mu^- + \mu^+$, where the spins of the outgoing particles are not specified. In lecture, we showed that the only nonzero contributions will be:

$$|\mathcal{M}_{RL \rightarrow XX}|^2 = |\mathcal{M}_{RL \rightarrow RL}|^2 + |\mathcal{M}_{RL \rightarrow LR}|^2$$

Compare your result to what we obtained in Homework 7:

$$\langle |\mathcal{M}|^2 \rangle = g_e^4 (1 + \cos^2 \theta)$$

Explain why your result makes sense physically.